Meteora

Every alarm, checked from the air.

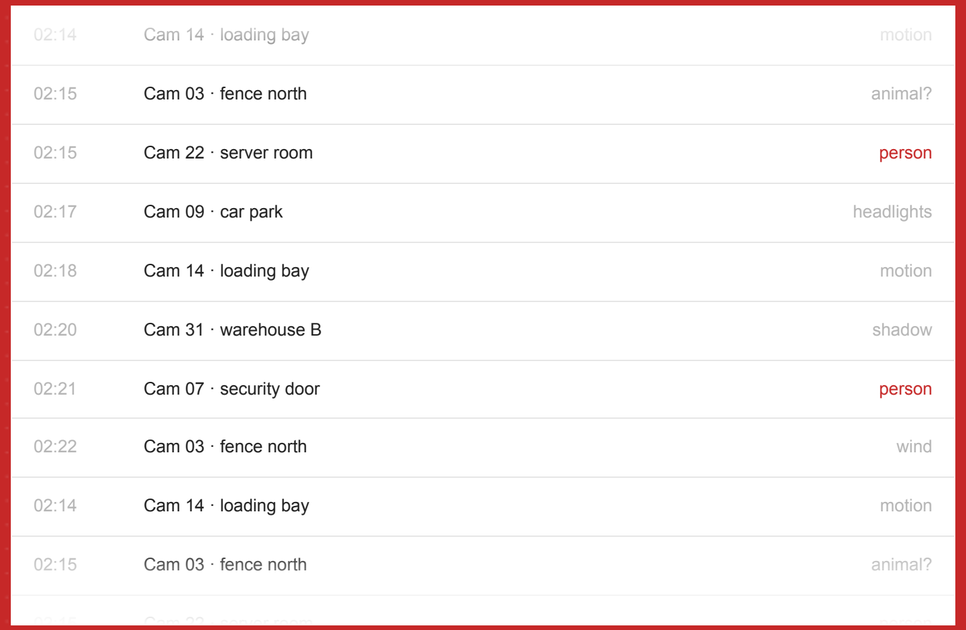
A camera spots someone where nobody should be. The nearest drone flies there, understands, and reports.

GROK BOT HACKATHON · PAVIA 2026 TRACK D · PROPOSED BY RIELLO

01 THE PROBLEM

A big site needs a closer look.

- 55,000 m² in one site, watched by cameras.
- Every alert needs a guard to walk over and check.
- Real intruders wait in the same queue as wind, shadows and cats.



02 OUR ANSWER

Send a drone to look first.

1



Detect

A camera sees a person inside a red zone.

2



Fly

The nearest drone takes off by itself and flies there.

3



Report

It understands the scene and tells the guard what is going on. People move only when it matters.

Anything that sees can be a detector.

TODAY

Phones
Any phone opens a link and watches for people inside the red zones you draw.

TOMORROW

The site's IP cameras
The cameras already on the walls.

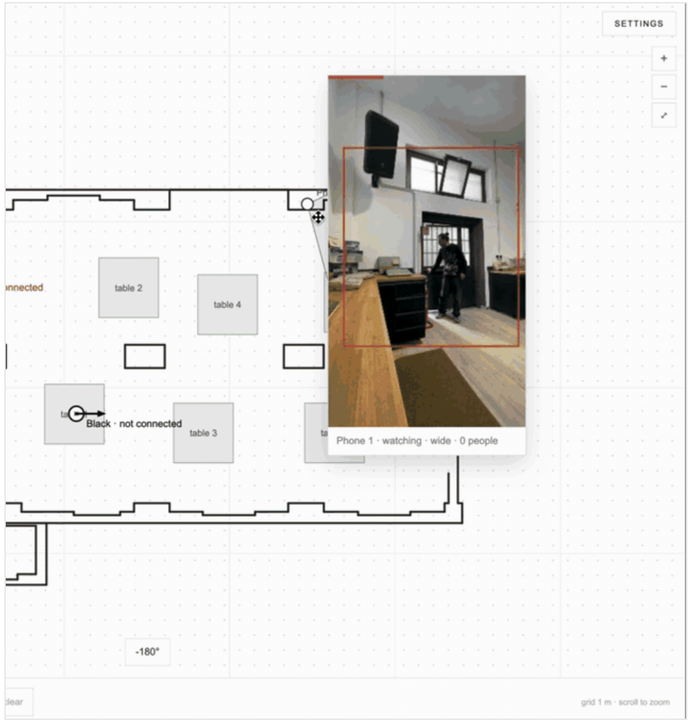
Door contacts
A door that opens at the wrong time.

Motion detectors
Movement in an empty room.

One alarm, whatever the source: the nearest drone goes to look.

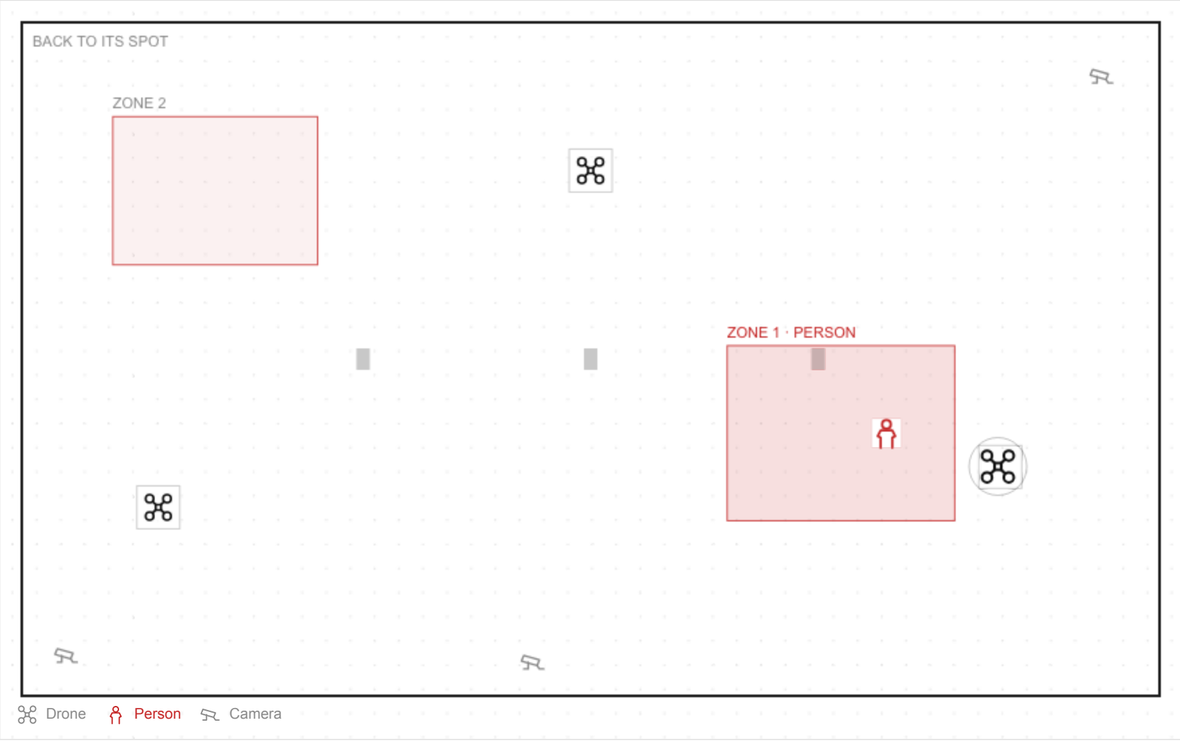


REAL The phone



REAL Zone 1 turns red

The nearest drone goes. By itself.

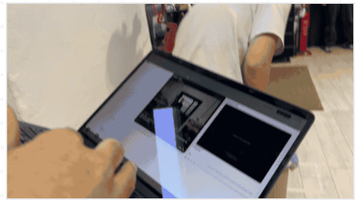


2 s from the alarm to take-off

0 people to dispatch: the system picks the nearest free drone

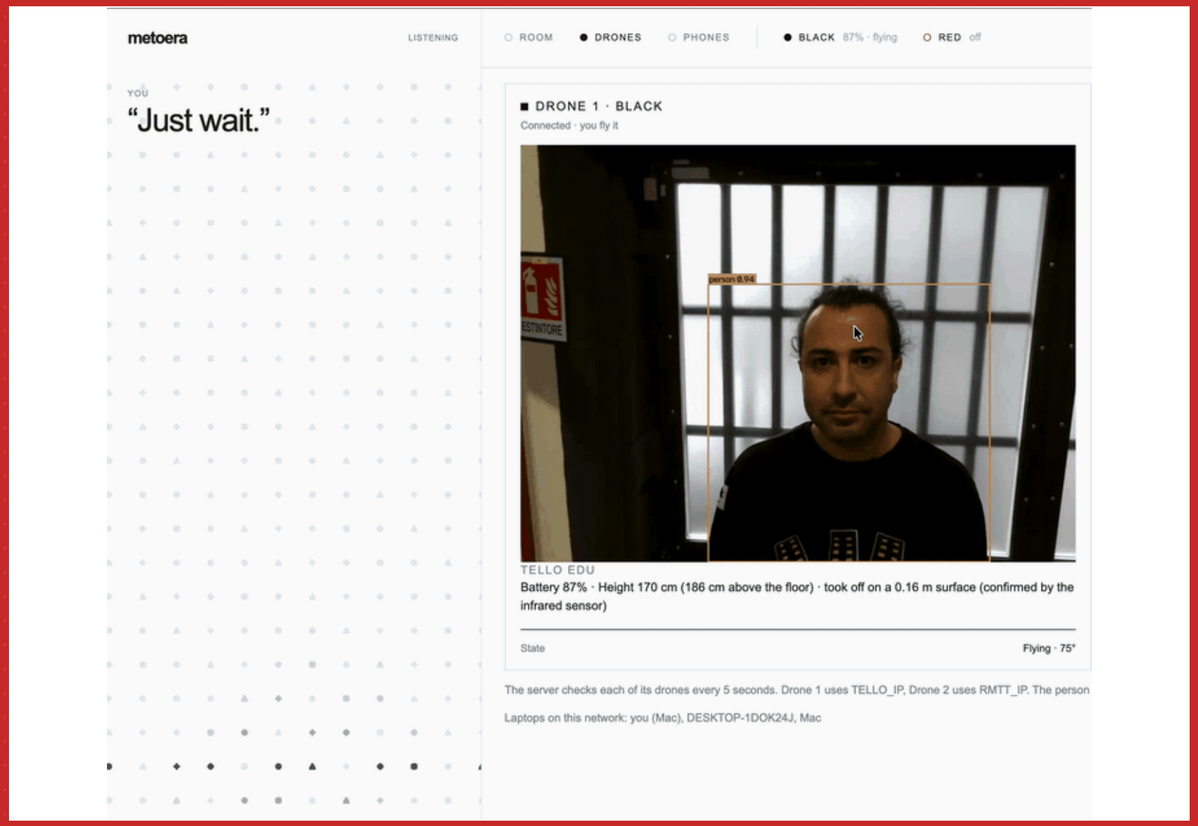
Home flies back to its spot when you say so; charging pads next

Times from a simulated mission on the real app.



REAL Black on its way, Capannone Gragnani

The drone reports.



REAL The drone camera, 3× speed.

3 photos: left, ahead, right | Person detector: 1 person, 0.94 | Grok vision writes it down

REPORT · ZONE 1 — CREATED BY METEORA'S AI · BLACK · 14:02

FOUND	1 person · confidence 0.94
WHO	A man stands in front of the drone, centre frame. Dark curly hair, short beard, black long-sleeve shirt with a graphic print.
WHERE	In front of a large window with vertical bars, next to a sign reading ESTINTORI.
NOT SAID	No names, no guesses about intentions: only what is in the photos.

Stay and watch | Go closer | Fly home

Talk to it like a colleague. Watch it think.

YOU
“Black, go to Zone 1. ”

GROK
On my way. Six metres, around pillar three.

YOU
“What do you see? ”

GROK
One person by the door, dark jacket, walking in.

YOU
“Stay and watch. ”

GROK
Holding here. I will tell you if anything changes.

WHAT GROK DOES

Understands
`go_to · drone Black · Zone 1`

Plans the route
`6 m, around pillar 3, 80 cm clear`

Takes off and flies
`checks the way ahead before each leg`

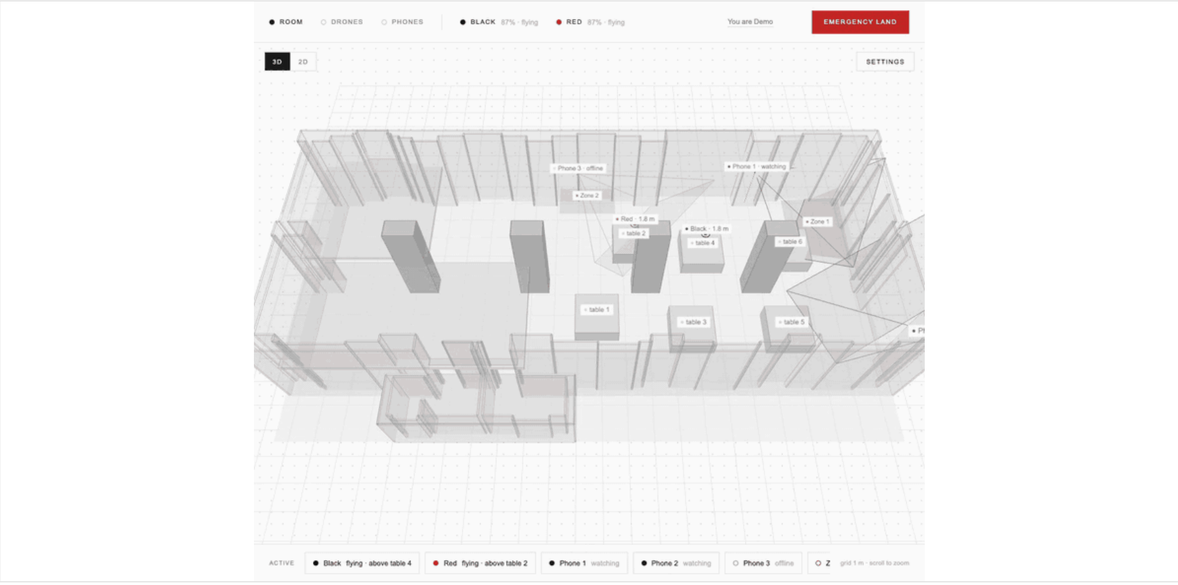
Inside Zone 1
`position confirmed on the plan`

Looks around
`3 photos: left, ahead, right`

Understands the scene
`1 person, described`

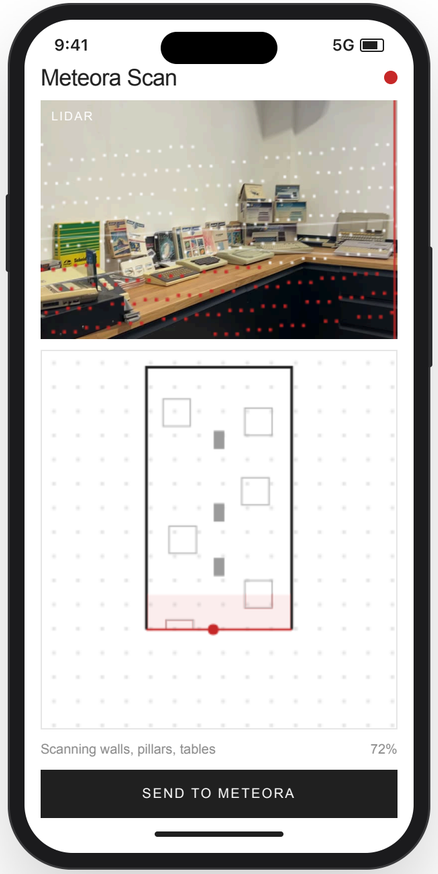
Waits for you
`hovering, keeps watching`

The site at real size.



Walls, pillars, tables, where every camera looks, the drones and the red zones. A red zone takes four clicks.

Walk the site once with a LiDAR phone: walls, pillars and tables appear at real size, and each camera phone finds where it stands and looks.



NEXT · METEORA SCAN

08 WHY GROK

One Grok runs the whole loop, and knows when to act.

Listens

Understands the operator in plain words.

Triages

Decides which events deserve a drone.

Understands

Turns what the drone sees into a report.

Alerts

Writes to security, and asks before escalating.

KNOWS WHEN TO ACT · SCENARIOS

02:21	Person at the server-room door, not on tonight’s list	SEND DRONE
21:05	Cleaner on the usual round, same path as every night	LOG ONLY
03:40	Camera 9 stopped sending pictures	ALERT MAINTENANCE
04:12	Drone battery at 20% on the way	FLY HOME

09 SPEED

Fast enough to check before anyone walks over.

ALARM TO EYES ON THE SPOT

22 s

Meteora	22 s
Guard, 300 m on foot	3.6 min

SIMULATED MISSION

GUARD: ESTIMATE, 1.4 M/S

NEXT DECISION IN A MISSION

0.6–1.1 s

Grok	≈1 s
Remote operator	≈3 s

VENUE OPERATOR: ESTIMATE

CHECK OF THE WAY AHEAD

0.11 s

Depth model	0.11 s
Human reaction	≈0.25 s

VENUE HUMAN: TYPICAL

KNOWS WHERE IT IS

85 %

Move accuracy	85%
Finds itself from 4 photos	2.8 s

CALIBRATION FLIGHT VENUE

People stay in charge.

Asks before acting

In the zone, Grok reports and asks: stay and watch, go closer, or fly home.

Emergency land

One button, always on screen. It cancels every command and lands every drone.

Lost signal, safe return

If the Wi-Fi drops, the drone backs up along its path and flies home, or lands after 12 s.

One pilot per drone

Whoever takes off flies it until it lands. Nobody else can command it.

Keeps its distance

80 cm from pillars, 50 cm from walls, and stops 60 cm before anything it sees.

Limits in code

Move sizes, battery and speed limits are enforced by the server, not left to the AI.

One kit, possibly live in an afternoon.

Plants

Warehouses

Data centres

Car parks

METEORA KIT

OFF THE SHELF

2 drones

Phones as cameras

1 laptop

Grok

From box to live in an afternoon.

Nothing to install on the phones, no new cameras, no wiring: the drones use the site Wi-Fi.

14:00	14:30	15:30	16:00	16:30
Drones on Wi-Fi	Scan the site	Phones in place	Red zones	Live

Museums

Campuses

Hospitals

Retail

Say it. The drones do it.

TAKE OFF AND LAND
“Black, take off.”

GO ANYWHERE ON THE PLAN
“Go to table 1, then table 2.”

CHECK A ZONE
“Go to Zone 1.”

TAKE A PHOTO
“Take a photo.”

DESCRIBE
“What do you see?”

LOOK ALL AROUND AND COUNT
“Do a 360 and count the people.”

FIND SOMEONE
“Find the person in the green shirt and go closer.”

REMEMBER A SPOT
“Mark this as position 1.”

GO BACK
“Go back to position 1.”

WHERE ARE YOU?
“Where is Black?”

BOTH DRONES AT ONCE
“Black to table 1, Red to table 5.”

CALIBRATE
“Run the calibration test.”

STOP, LAND
“Land now.”

Meteora

Look first, then alert.

Grok Bot Hackathon · Pavia 2026 · Track D, proposed by Riello

TEAM

[Team names]

CODE

github.com/arn9ve/Meteora